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## Biopol4motive

# Project Overview

Autoneum. Mastering sound and heat.

## Biopol4motive

# Project in a glance

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- **Project target:**
  *Create two grades of flexible PU foams (i.e. high-resilience and visco-elastic) with a bio-content of 35-40 % for automotive acoustical applications*
- Consortium between Oleon, Huntsman, and Autoneum managed by the Flemish association *"Catalisti"*
- Project start date: *1st of April*
  Duration: *33 months*

Prolonged 2x 6 month
New end 31.12.2024

CATALISTI
WE MEAN BUSINESS

oleon
a natural chemistry

HUNTSMAN

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| Organisation | Contact person | Contact details (phone, email) |
|---|---|---|
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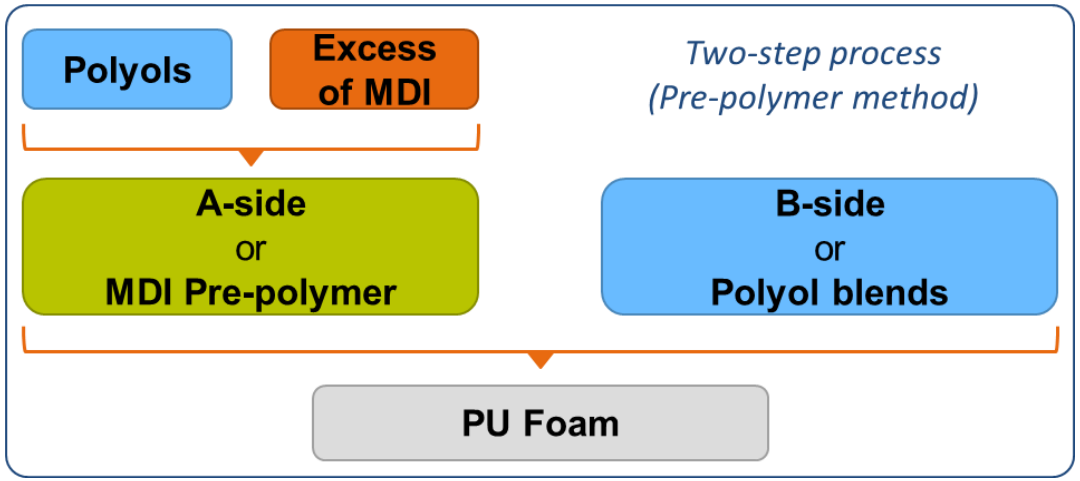
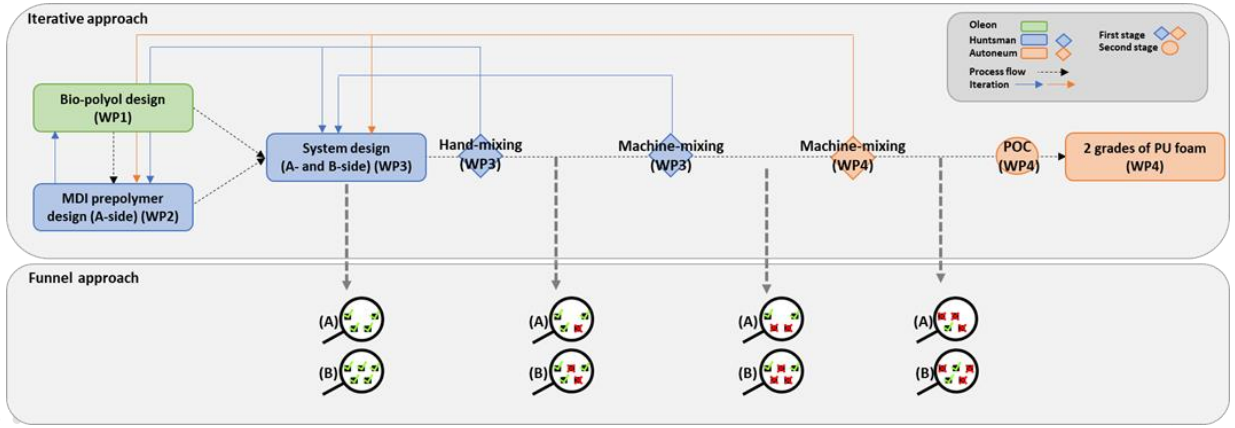
| **Autoneum** | Maurizio Mantovani | +41 52 244 84 89<br>maurizio.mantovani@autoneum.com |

# Technical overview



- BIOPOL4MOTIVE approach is to synthesize Flexible foams via a **two-step process (pre-polymer method)**.
- To achieve 25% sustainable content on total part level (Carpet) the PU foam must reach **35-40% in weight of bio-content**.
- To achieved the required bio-content the focus will be the "fit for purpose" design of:
  - **4 bio-polyol** for the isocyanate pre-polymer **(A-side)**
  - **5 bio-polyol** for the polyol blend **(B-side)**
- A **funnel approach** is used to down-select the three most promising ones:
  - 1 bio-polyol for the **(A-side) for both grades**
  - 1 bio-polyol for the **(B-side) for VEF grade**
  - 1 bio-polyol for the **(B-side) for HR grade**



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# Concrete Targets

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**Bio-polyol** concrete targets:

| Parameter | Value |
|---|---|
| Molecular weight ($M_w$) | 3500-6000 g/mol |
| Functionality | at least 2.1, more preferably close to 3 |
| Hydroxyl value ($OH_v$) | 28 - 48 mg KOH/g |
| Low residual acid value | 2 mg KOH/g |
| Primary hydroxyl groups | above 80% |
| Soft block length | at least 100 atoms |
| Viscosity at 25°C | < 2000 cP |
| Sodium and Potassium | < 5 ppm (A-side) |
| Water content | ≤ 0.1% |
| Bio-content | >50% |
| biodegradability | 75% within incubation time of 28 days |

- Polyether-polyester based
- A bio-polyol which does not compete with the food chain
- REACh registered when substance, or classed as polymer

**MDI pre-polymers (A-side)**, concrete targets:

| Parameter | Value |
|---|---|
| Pre-polymer functionality (Fn) | 2.15-2.30 |
| Reactive isocyanate groups in the A-side | around 18% |
| Viscosity | < 3500 Cps<br>< 2000 Cps at operating temperature |
| Acidity | 0.004% |
| Dimer content | <350 ppm |
| Hydrolysable chlorine | <350 ppm |
| Storage stability | pre-polymer lifetime of 6 months at 20°C |

- Develop a theoretical model for pre-polymerisation of bio-polyols that covers the relationship between the pre-polymer composition, characteristics, processing behavior and properties of a PU foam.
- Develop a model to predict the pre-polymer storage stability.

**PU Foam** concrete targets:

| Parameter | Value |
|---|---|
| compression hardness (both grades) | 7-10 kpa |
| Loss Factor HR | 0.20 – 0.30 % |
| Loss Factor VEF | 0.33 – 0.43 % |
| OEM speification HR | Volvo Technical regulation |
| OEM speification VEF | Daimler DBL5452 |

- To develop a theoretical model for bio-based polymer synthesis that relates the composition of the MDI oligomer mixture, total MDI amount in the foam, amount of water, B-side polyol(s) backbone structure and overall PU polymer soft block length.

# Project workflow

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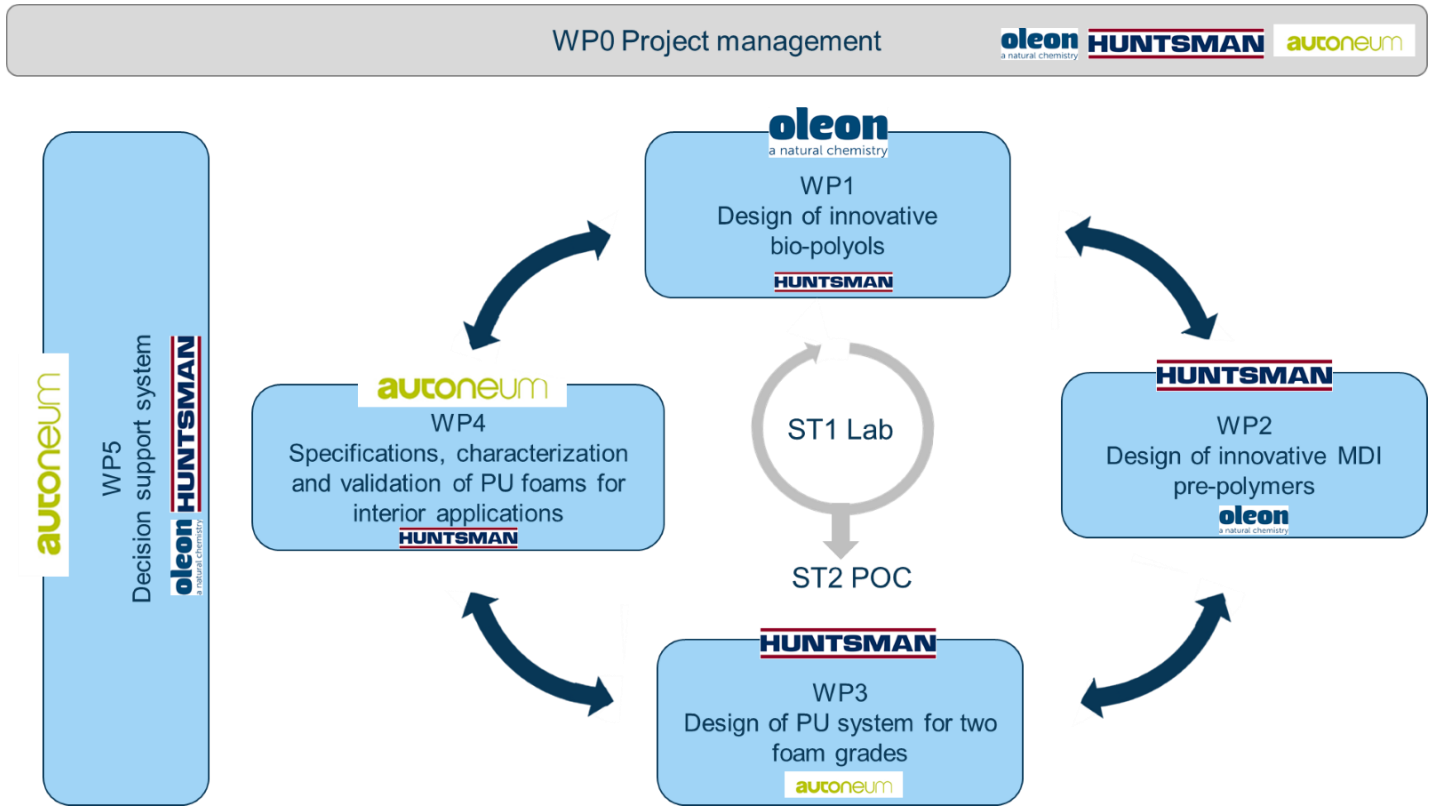
***Remarks:***

- The work is organized in five highly interacting work packages
- WP1 to WP4 are the core of the research project




*ST1 Lab: Stage 1 Lab-scale* *ST2 POC: Stage 2 Proof of concept*

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# Oleon Polyol Technology

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- Base is rapeseed oil
- the oil is derived as a side product from animal food production (rapeseed fibers)
  - ➢ No competition with food chain

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# Project steps – in short

- Production of multiple (28 novel) base polyols at Oleon
- Evaluation of 15 novel bio polyols at Huntsman on both the iso and the polyol-side
- Several screening trials in Huntsman's lab via handmixing
- Improvement of base polyols and concentration on finally 2 products
- Improvement of production process to get repeatable results
- Machine evaluation showed narrow processing range
- Trials in Autoneum showed big demoldability issues (tearing at demold and when also parts are handled)
- Multiple alignments on LCA calculation
- Big price increase for rapeseed oil polyol
- Huntsman brought alternative solutions, which also suffered from tearing at demold, one even more
- Agreed to reduce bio-content from 40% to about 20% to make production feasible (no content from Oleon)

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## Biopol4Motive

# Trial of w33/2024 with HRB 30510 / Suprasec 2616 in Winterthur

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# Global data

Materials:

- Acoustiflex HRB 30510 BX1 (MU240724) Polyol
- Suprasec 2616 Isocyanate
- Ewomold 5211 release agent
- Additional catalyst: Additive 9813

Tools:

- Flat tool: 1400 mm x 1200 mm x 20 mm
- MFA2 LHD front

Tool temperature: 63-65°C

Raw material temperature: 35-40°C

# Trial w33/2024 – flat tool

- Trials run in Isokell size flat tool at 20mm thickness
- 2 different mixing ratios (I/P): 0.62 and 0.66
- Material filled the tool at both targeted densities of 50 and 45 kg/m3
- Foam tearing is improved versus the earlier formulation development status
- Foam is open, does not show signs of shrinkage
- Foam requires a tight tool, needed to work at 300 kPa (30to) clamping force
- Material requires a lot of release agent compared to standard formula; the required amount is approximately 2-fold

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# Trial w33/2024 – MFA2 LHD tool

A) Original formulation to sensitive for carpet tool

- did not fill completely at 2000g shotweight (2200g = 60g/L)
- Clamping force set to 500 kPa (50to) !!

B) modification 1: Acoustiflex HRB 30510 BX1 (MU240724) +0.3% catalyst Additive 9813

- formulation is more stable, still it requires good tool sealing
- lower mixing 0.62 (I/P) ratio still very sensitive on carpet, filling at ca. 2000g (2200g used for calculated 60g/L density)
- higher mixing ratio 0.66 provides more stability, could go down to 1900 g and borderline 1835g shot-weight (calculated 50g/L)
- should check static hardness and dynamic Young's modulus on carpet level
- requires a very high amount of release agent (about 2-times the amount vs. standard HR system)
- demolded after 60s, which was still very challenging
- **Agreed with plant to run in production in order to get an indication where we are vs standard foam**

C) modification 2: added another 0.3% additive 9813 on top of modification 1

- foam more stable
- strong issues to fill the tool
- no good parts were produced, did not fill at 1900g
- foam might be too fast now
- free rise bucket shot in 10L bucket: EOB 10s, full collapse at 14s

# Trial w33/2024 – MFA2 LHD tool

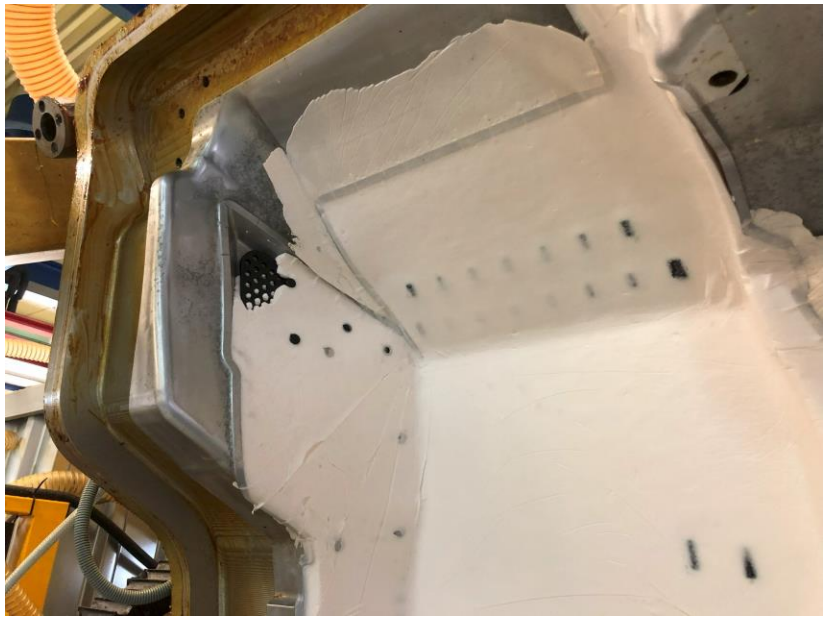
Good parts produced:
Parts -10 with 2200g shows some overfill issue in the passenger side crash pad area (just above the naming) and looks a bit blown.
With less shot-weight that area becomes more flat

# Trial w33/2024 – MFA2 LHD tool – shortshots on foil

We produced parts with different shotweights. The carpet was separated from the foam by a foil

Matthew will do the required Foamat tests for the simulation.

1900 g



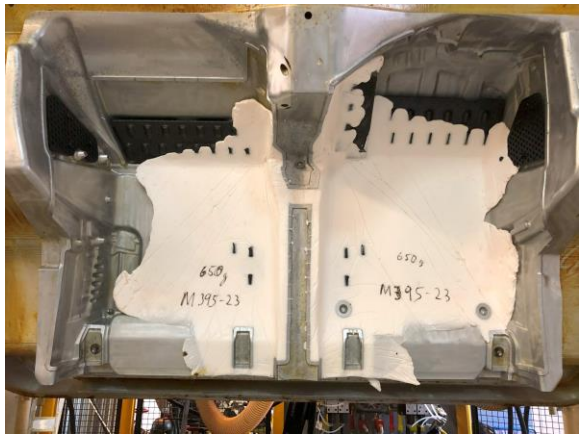
# Trial w33/2024 – MFA2 LHD tool – shortshots on foil

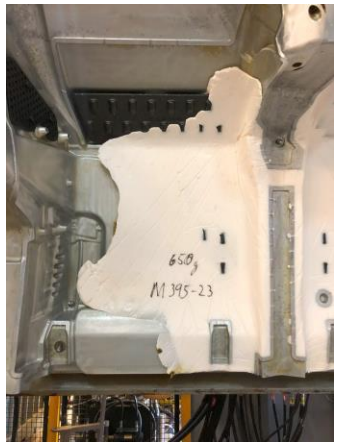
1300g

With blocked vent

With clean vent

# Trial w33/2024 – MFA2 LHD tool – shortshots on foil



650 g

Injection channel design:

cross section on driver side is much larger

# Part and tool ID

(part numbers A177 680 2000 and A177 6801900 produced in 1 tool with 1 common shot;

tool maker: Autoneum Hnatnice

tool number: 17116/525 (as far as I could recognize)); tool number 12

# Agreed next steps

| | |
|---|---|
| • Get a time slot for running modification B in Genk | Wolfgang with Herman Sanen |
| • Run FOAMAT on mod B and C and give data to Michael | done, Matthew |
| • Update general status on simulation | Valerio |